Abstract algebra – 1

Finite groups: counting

Homework

13 сентября 2026 года

Theoretical minimum. A group is a set with an associative operation, a neutral element and inverses; the order $\operatorname{ord}(x)$ of an element is the least $k > 0$ with $x^k = e$; the index $[G : H]$ is the number of cosets xH. Lagrange's theorem says that the cosets partition the group, so $|G| = [G : H]\,|H|$, whence $\operatorname{ord}(x)$ divides $|G|$ and $x^{|G|} = e$. Cauchy's theorem says that if a prime p divides $|G|$ then G has an element of order p; for $p = 2$ it follows from the count $\#\{x : x^2 = e\} \equiv |G| \pmod 2$, obtained by pairing x with x^{-1}. The class equation $|G| = |\operatorname{Z}(G)| + \sum[G : C(x_i)]$ shows that a group of order p^k has a non-trivial centre; if $G/\operatorname{Z}(G)$ is cyclic then G is abelian. Every finite abelian group is a direct sum of cyclic groups $\mathbb{Z}/d_1 \oplus \cdots \oplus \mathbb{Z}/d_k$ with $d_1 \mid \cdots \mid d_k$, and d_k is the exponent; a group with $x^2 = e$ for all x is isomorphic to $(\mathbb{Z}/2)^k$.

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1. Let r, s, t be pairwise coprime positive integers and let a, b be elements of a commutative group with $a^r = b^s = (ab)^t = e$. Prove that $a = b = e$. Does the same conclusion hold in an arbitrary non-commutative group? (IMC)

2. Prove that a group G in which exactly two elements other than the identity commute with each other is isomorphic to $\mathbb{Z}/3\mathbb{Z}$ or to S_3. (Румыния, 1996)

3. Let G be a finite group of order n, let $H = \{x \in G : x^2 = e\}$ and $p = |H|$. Prove that (a) $|H \cap xH| \geqslant 2p - n$ for every $x \in G$; (b) if $p > \frac{3n}{4}$ then G is commutative; (c) if $\frac{n}{2} < p \leqslant \frac{3n}{4}$ then G is not commutative. (Румыния, 2010)

4. For a finite group G let $n(G)$ be the number of its elements and $s(G)$ the number of its subgroups. Decide whether each of the following is

true: (a) for every $a > 0$ there is a finite group with $n(G)/s(G) < a$; (b) for every $a > 0$ there is a finite group with $n(G)/s(G) > a$.
(Румыния, 1999)

5. Let G be a group with m elements and H a proper subgroup with $n \geqslant 2$ elements. Put $H^x = \{xhx^{-1} : h \in H\}$ and assume $H^x \cap H = \{e\}$ for every $x \in G \setminus H$. Prove that $H^x = H^y$ if and only if $x^{-1}y \in H$, and find $\left|\bigcup_{x \in G} H^x\right|$ as a function of m and n. (Румыния, 2005)

6. Let $*$ be a commutative and associative binary operation on a set S such that for all $x, y \in S$ there exists $z \in S$ with $x * z = y$. Show that $a * c = b * c$ implies $a = b$. (Putnam 2012, A2)

7. Prove that if $G/\operatorname{Z}(G)$ is cyclic then G is abelian, and deduce that every group of order p^2 is abelian. Then show that for every prime p there is a non-abelian group of order p^3, so the statement is sharp.

8. Let $\pi\colon G \to Q$ be a surjective homomorphism of finite groups. Prove that there exists a subgroup $H \leqslant G$ such that $\pi(H) = Q$ and the set of primes dividing $|H|$ coincides with the set of primes dividing $|Q|$.
(IMC 2025, shortlist)

9. Let n be a squarefree positive integer and G a group of order n. Let d be the largest order of an element of G and N the number of elements of order d. Prove that (a) N is divisible by n/d; (b) $N^2 \geqslant \varphi(n)$.
(Румыния, 2026)

10. Let G be a finite group with the property that for every automorphism f of G there exists $m \in \mathbb{N}$ with $f(x) = x^m$ for all $x \in G$. Prove that G is commutative. (Румыния, 2017)